

TIME: 25 Minutes Total Marks: 20

Student_ID:

1. “Any set of three points or any two lines on a plane is always sufficient to uniquely determine the equation of that plane.” - Assess the correctness of the statement. 4
2. In a 3D world, two cats, Milo and Tilo, are playfully chasing each other. At one moment, Milo is located at the point $M = (-6, 3, 5)$. Suddenly, an infinite vertical wall appears along the plane $x = 4$, dividing the space into two separate regions. Determine the set of all possible coordinates for Tilo, $T = (x, y, z)$, such that they remain separated by the wall. 3
3. A man and a gunman are both initially located on the XY -plane of a 3D coordinate system. The man is standing at the origin and his height is 5 units extending along the positive Z -axis. 13
As the gunman begins running directly towards the man, the man simultaneously starts running along the positive Y -axis. Suddenly, the gunman fires a bullet in the direction of the vector $(-4, 8, 2.5)$. By that time the man had already covered a distance of 8 units along the Y -axis with his body tilted forward at an angle of 60° from the vertical (Z -axis), bending toward the Y -axis. The bullet narrowly misses the man. However, if the man had been standing upright, the bullet would have struck him exactly at the midpoint of his body i.e., 2.5 units above the XY plane.

Determine the exact position from which the bullet was fired.

